

Ukuqala Patient API

Comprehensive Technical Specification & System Documentation

1. Executive Summary

This document serves as the definitive technical record for the **Ukuqala Patient API**. It covers the system architecture, security protocols, database design, and API endpoint specifications. The system is designed to provide medical professionals with a secure, scalable, and efficient way to manage patient health records.

"The transition to UUID v4 and Version 4 (v4) routing ensures that the system is resilient against data collisions and follows industry-standard security practices for healthcare software."

2. Data Architecture & Design Logic

The system utilizes a relational database model implemented via **Sequelize ORM** and **PostgreSQL**. Identity management has been migrated from sequential integers to Universally Unique Identifiers (UUIDs).

2.1 Class Diagram (Logic Layer)

User Object (The Practitioner)

- Attributes: UUID (ID), String (Name), String (Email/Unique), String (Hash)
- Hooks: beforeCreate (Bcrypt password salting/hashing)

(1 User) 1 ----- * (N Patients)

Patient Object (The Record)

- Attributes: UUID (ID), String (Name), Integer (Age), String (Condition), UUID (userId)
- Logic: Strict NOT NULL constraints on all medical fields.

2.2 Database Schema Specification

Table	Attribute	Data Type	Constraint
Users	id	UUID	PRIMARY KEY
Users	email	VARCHAR	UNIQUE, NOT NULL
Patients	id	UUID	PRIMARY KEY
Patients	userId	UUID	FOREIGN KEY -> Users(id)
Patients	age	INTEGER	NOT NULL

3. API Implementation (v4)

3.1 Authentication Strategy

Security is handled via **JSON Web Tokens (JWT)**. Upon successful login, the system issues a Bearer token which must be included in the Authorization header of all subsequent requests.

3.2 Endpoints Table

Module	Method	Endpoint	Description
Auth	POST	/api/v4/auth/register	User creation with encrypted password
Auth	POST	/api/v4/auth/login	

Module	Method	Endpoint	Description
			Credential verification & Token issuance
Patient	POST	/api/v4/patients	Add record (Requires Token)
Patient	GET	/api/v4/patients	List all records owned by user
Patient	PUT	/api/v4/patients/:id	Update record by UUID
Patient	DELETE	/api/v4/patients/:id	Remove record from database

4. Engineering Improvements & Debugging

The following technical obstacles were successfully cleared to reach the current stable state:

- **JSON Payload Mapping:** Fixed notNull Violation by synchronizing model definitions with name, age, and condition inputs.
- **Middleware Logic:** Corrected TypeError: argument handler must be a function by refining the module.exports pattern in the controller layer.
- **Concurrency Safety:** Implemented UUID v4 to prevent primary key overlap during simultaneous record creation.

5. Installation & Deployment

The source code is managed via Git and synchronized with the master branch on GitHub.

```
git clone https://github.com/SirriKamaric/ukuqala-patient-api.git npm install #
Set DB credentials in .env npm start
```

Official System Documentation - Ukuqala Project

© 2026 Technology Documentation Division